

Notes: Bybee, Kelly, Manela, and Xiu (JF, 2024)

Business News and Business Cycles

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1 Big picture

- **Question.** Can business news text measure the state of the economy and help forecast macroeconomic and financial dynamics?
- **Core input.** The authors use the full text of roughly 800,000 *Wall Street Journal* articles from 1984 to 2017.
- **Core method.** They estimate a latent Dirichlet allocation (LDA) topic model that maps ultrahigh-dimensional newspaper text into interpretable news topics and monthly topic-attention series.
- **Main measurement result.** Business news decomposes into interpretable themes such as “recession,” “oil market,” “health insurance,” “IPOs,” “takeovers,” and “options/VIX.”
- **Model-selection result.** A 180-topic specification is chosen as an approximately optimal balance between fit and parsimony.
- **Reconstructing economic series.** Small active sets of news topics reconstruct many macro-finance time series: industrial production, employment, stock returns, volatility, IPO volume, LBO volume, bankruptcies, CDS spreads, and industry volatility.
- **Most important topic.** “Recession” attention is repeatedly selected as a key explanatory and predictive variable for macroeconomic dynamics and stock market outcomes.
- **VAR result.** A shock to recession news attention predicts a large decline in industrial production and employment, even after controlling for standard macro-finance variables.
- **Interpretability result.** Topic models allow “narrative retrieval”: the researcher can trace a model forecast back to the specific articles and narratives that drive it.
- **Market timing result.** LDA-based news attention improves out-of-sample market-timing performance relative to buy-and-hold, EPU, word-count recession signals, and Welch–Goyal predictors.
- **Economic takeaway.** News text is not just noisy commentary on the economy. It is a high-dimensional, interpretable, and forecast-relevant measurement system for business-cycle states.

2 Data and measurement

2.1 WSJ corpus

- The sample is the full text of *Wall Street Journal* articles from January 1984 through June 2017.
- The authors restrict to core WSJ sections to reduce changes in newspaper composition over time.
- The raw corpus is transformed into an article-term count matrix.
- Preprocessing includes term filtering and lemmatization to reduce vocabulary noise.

Interpretation: The paper treats the newspaper as an information intermediary that converts economic events, expectations, and interpretations into observable text.

2.2 Bag-of-words representation

Let w be the article-term matrix. It has dimension $T \times V$, where T is the number of articles and V is the number of unique terms. The element

$$w_{t,v}$$

counts how often term v appears in article t .

Interpretation: Bag-of-words discards grammar and word order but keeps the information needed to estimate topical structure across a very large corpus.

2.3 Topic attention

For each article t , LDA estimates topic weights

$$\hat{\theta}_t = (\hat{\theta}_{t,1}, \dots, \hat{\theta}_{t,K}), \quad \sum_{k=1}^K \hat{\theta}_{t,k} = 1.$$

Monthly attention to topic k is the average topic weight across articles published in month m :

$$\hat{\Theta}_{m,k} = \frac{1}{N_m} \sum_{t \in m} \hat{\theta}_{t,k}.$$

Interpretation: A topic-attention series is a monthly quantitative index of how much the newspaper is discussing a particular economic narrative.

2.4 Topic interpretation

Each topic k is a probability distribution over terms:

$$\hat{\phi}_k = (\hat{\phi}_{k,1}, \dots, \hat{\phi}_{k,V}), \quad \sum_{v=1}^V \hat{\phi}_{k,v} = 1.$$

The highest-probability terms provide a human-readable label.

Interpretation: A topic such as “recession” is not hand-coded. It emerges as a cluster of words like downturn, slowdown, weak economy, recession, job cuts, and related terms.

2.5 Macroeconomic and financial variables

The paper links topic attention to:

- industrial production growth;
- employment growth;
- aggregate stock market returns;
- stock market volatility;
- IPO and LBO dollar volumes;
- bankruptcy filings;
- European sovereign CDS spreads;
- industry-level volatility innovations;
- market-timing returns.

Interpretation: The empirical question is whether interpretable text-derived variables track and forecast conventional numerical indicators.

3 Models and empirical specifications

3.1 LDA model

LDA assumes the article-level term-count vector is multinomial:

$$w_t \sim \text{Mult}(\Phi^\top \theta_t, N_t),$$

where:

- N_t is article length;
- $\Phi = [\phi_1, \dots, \phi_K]^\top$ is the topic-term matrix;
- θ_t is article t 's allocation of attention across topics.

Interpretation: LDA is a factor model for text: topics are common factors, and article-specific attention weights are factor loadings.

3.2 Choosing the number of topics

The authors estimate candidate LDA models for K ranging from 50 to 250. Model fit is evaluated using Bayes factors and cross-validation:

$$K^* \approx \arg \max_K \{\text{model fit}(K)\}.$$

The chosen specification uses 180 topics.

Interpretation: Too few topics merge distinct narratives into broad buckets. Too many topics overfit and make the news map harder to interpret.

3.3 Active-set lasso reconstruction

For an economic time series y_t , the paper estimates sparse regressions of the form

$$y_t = a + \sum_{k \in S_y} \beta_k \hat{\Theta}_{t,k} + u_t,$$

or, for some outcomes, using innovations or transformations of topic attention and the dependent variable. The active set S_y is chosen by lasso, often restricted to five regressors for interpretability. Inference uses postselection adjustments.

Interpretation: The goal is not only prediction. The selected topics provide a concise narrative interpretation of each macro-finance series.

3.4 Text-augmented VAR

The baseline macro VAR augments a standard BBD-style macro-finance VAR with recession attention:

$$Y_t = c + \sum_{\ell=1}^p A_\ell Y_{t-\ell} + \varepsilon_t,$$

where Y_t contains recession attention and standard macro-finance variables such as stock prices, interest rates, employment, and industrial production.

A news impulse is defined as a shift in recession attention from its 5th percentile to its 95th percentile.

Interpretation: This design asks whether news text contains predictive information beyond standard numerical macroeconomic variables.

3.5 Impulse responses

The paper computes impulse responses:

$$IRF_h = \frac{\partial Y_{t+h}}{\partial \varepsilon_t^{\text{recession}}}.$$

The main outcomes are industrial production and employment.

Interpretation: A recession-attention shock is not interpreted as a clean causal shock. It is a shock to the measured narrative state of the economy that contains incremental forecasting information.

3.6 Group-lasso VAR topic selection

Because the VAR cannot include all 180 topics, the authors use group lasso:

$$\min_A \sum_t \left\| Y_t - c - \sum_{\ell=1}^p A_\ell X_{t-\ell} \right\|^2 + \lambda \sum_g \|A_g\|_2.$$

The selected topic at the cross-validated penalty is “recession.”

Interpretation: This validates the baseline choice of recession attention as the news topic most useful for macroeconomic VAR prediction.

3.7 Narrative retrieval

For a topic k in month t , the representative article is

$$i_{t,k}^* = \arg \max_{i \in t} \hat{\theta}_{i,k}.$$

The method can trace model-implied forecast movements back to the articles most representative of the relevant topic.

Interpretation: Narrative retrieval makes the machine-learning model interpretable. It points the researcher to the articles behind a forecast instead of leaving the topic model as a black box.

3.8 Market timing with news attention

For return forecasting, the model estimates rolling predictive regressions:

$$r_{t+1} = a_t + \sum_{k \in S_t} \beta_{t,k} \hat{\Theta}_{t,k} + u_{t+1}.$$

The strategy holds the predicted market return:

$$R_{t+1}^{\text{strategy}} = \hat{r}_{t+1} r_{t+1}.$$

For narrative retrieval of market-timing gains, the most relevant topic is

$$\hat{k}_{t+1} = \arg \max_k \hat{\Theta}_{t,k} \hat{\beta}_{t,k} r_{t+1}.$$

Interpretation: The topic model is used not just to describe business cycles, but also to form out-of-sample return forecasts.

4 Identification and empirical design

4.1 What identifies the paper’s empirical contribution

- The paper is not a causal identification design.
- The main evidence comes from measurement, reconstruction, forecasting, and robustness exercises.
- News topics are learned without hand-coding the specific outcomes.
- The same topic-attention series are then tested against many external macro-finance variables.

Interpretation: Credibility comes from out-of-sample fit, interpretability, robustness, and the ability to link numerical forecasts back to article narratives.

4.2 Avoiding look-ahead bias

- Full-sample LDA uses all articles to estimate topics.
- For forecasting and robustness, the paper also uses online LDA (oLDA), which estimates attention without using future article information.
- The oLDA results remain close to the full-sample LDA results in the VAR and market-timing analyses.

Interpretation: This matters because a text model estimated using the full sample could otherwise encode information from the future.

4.3 Why the paper emphasizes interpretability

- Topic labels and key terms explain what each time series measures.
- Lasso active sets provide a short list of topics associated with each outcome.
- Narrative retrieval finds the articles that drive model forecasts.

Interpretation: The paper combines predictive machine learning with economic reading. The model is useful because it is both quantitative and narratively interpretable.

4.4 Limitations

- News attention does not by itself establish causality.
- The WSJ is one newspaper and reflects editorial choices, reader demand, and news production technology.
- Topic models depend on preprocessing choices and the number of topics.
- Some selected topics may reflect anticipated effects rather than causes.
- Long-run comparisons use front-page abstracts for early decades, not full article text.

Interpretation: The paper's claims are strongest as measurement and forecasting claims, not as structural causal statements about media effects.

5 Tables and figures

This section groups adjacent original tables and figures to save space and speed up reading. All visuals are cropped directly from the paper.

5.1 Figure 1 + Figure 3: corpus size, model selection, and topic examples

Read:

- Figure 1 shows the number of processed WSJ terms and articles over time and the model-selection evidence across candidate topic counts.
- Bayes-factor and cross-validation evidence point to a 170–180 topic specification.
- Figure 3 shows six topic-attention series and their key terms: recession, health insurance, elections, earnings forecasts, terrorism, and natural disasters.
- Topics are persistent, recurrent, seasonal, or emergent.

Economic message: The topic model turns a very large newspaper corpus into a finite menu of interpretable narrative state variables.

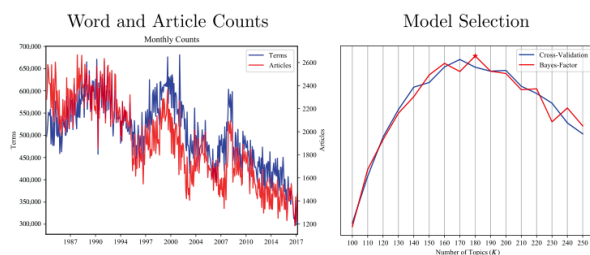


Figure 1. WSJ observation counts and model selection. The left panel plots the postprocessing article count and the total term counts aggregated over all articles each month. The right panel plots the cross-validated model fit (average log-likelihood over 10 cross-validation folds) and the Bayes factor for models with topic count K ranging from 50 to 250 in increments of 10 (we omit labels on the right panel vertical axis as the model fit comparisons are relative and specific values are uninformative). (Color figure can be viewed at wileyonlinelibrary.com)

Figure 1: counts and model selection

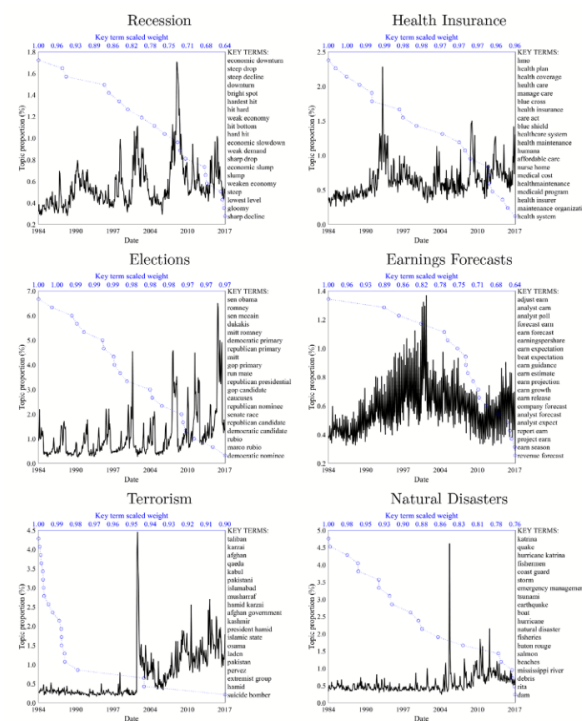


Figure 3. Topic attention examples. The black line shows topic attention as a percent of

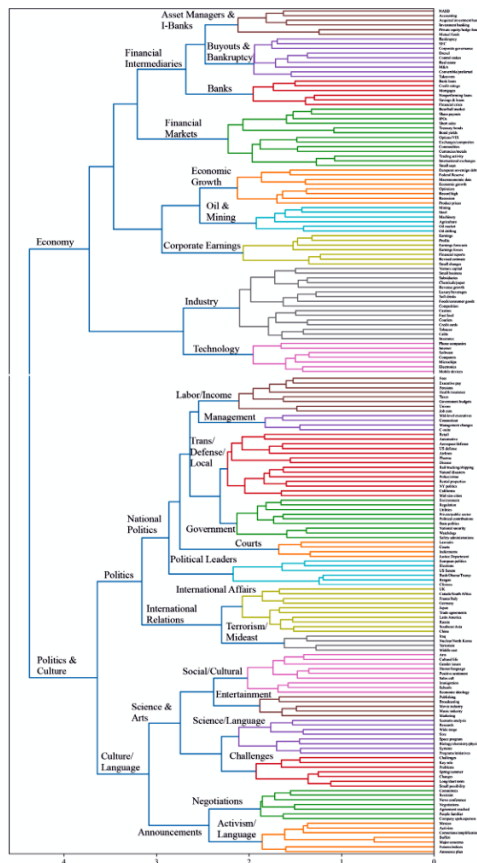
Figure 3: topic attention examples

5.2 Figure 2: hierarchical taxonomy of business news

Read:

- Figure 2 clusters the 180 topics into broader metatopics.
- The top branches split economy-related news from politics/culture-related news.
- Within economy news, major groups include financial intermediaries, financial markets, economic growth, oil and mining, corporate earnings, industry, and technology.
- The hierarchy shows that topic modeling produces a coherent map of business news.

Economic message: The taxonomy is evidence that the model does not simply create arbitrary statistical factors; it creates an interpretable news ontology.



5.3 Table I + Table II: reconstructing macro series and stock returns

Read:

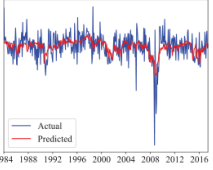
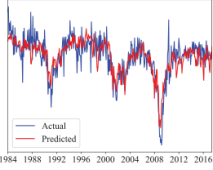
- Table I shows that five news topics can reconstruct industrial production growth, employment growth, market returns, and market volatility.
- Recession attention enters negatively for industrial production, employment, and market returns, and positively for market volatility.
- The fitted R^2 values are large: 0.21 for industrial production, 0.59 for employment, 0.25 for market returns, and 0.63 for market volatility.
- Table II shows that news topics explain stock returns far better than CFNAI or FRED-MD benchmark variables in the same sparse framework.

Economic message: News attention is a surprisingly strong quantitative summary of both macroeconomic conditions and financial market states.

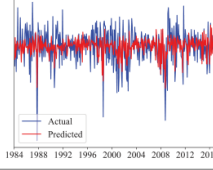
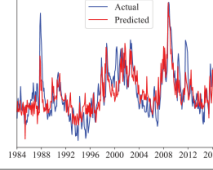
Table I
Reconstructing Macroeconomic Time Series

Five-regressor active-set regression estimates based on lasso selection with p -values adjusted for postselection inference.

Industrial Production Growth			Employment Growth		
Topic	Coeff.	p -Val.	Topic	Coeff.	p -Val.
Recession	-0.38	0.00	Recession	-0.61	0.00
Oil market	-0.17	0.00	Rail/trucking/shipping	0.22	0.01
Southeast Asia	0.11	0.10	Bush/Obama/Trump	-0.15	0.09
Health insurance	0.06	0.93	Iraq	-0.14	0.01
Clintons	0.03	0.40	Clintons	0.12	0.01
R^2	0.21		R^2	0.59	

Market Returns			Market Volatility		
Topic	Coeff.	p -Val.	Topic	Coeff.	p -Val.
Recession	-0.35	0.00	Recession	0.48	0.00
Problems	-0.19	0.00	Options/VIX	0.24	0.00
Convertible/preferred	-0.16	0.21	Problems	0.22	0.00
Record high	0.12	0.08	Small business	-0.18	0.00
Options/VIX	-0.11	0.76	Investment banking	0.13	0.00
R^2	0.25		R^2	0.63	

is the lowest determinant of market returns with a coefficient of -0.35. The
Table I: reconstructing macroeconomic time series

Table II
Reconstructing Stock Market Returns

Five-regressor active-set regression estimates based on lasso selection with p -values adjusted for postselection inference. Full R^2 corresponds to the fit with both topics and benchmark variables. Benchmark R^2 corresponds to the fit with only benchmark variables.

Topic Model & CFNAI			Topic Model & FRED-MD		
Topic	Coeff.	p -Val.	Topic	Coeff.	p -Val.
Recession	-0.35	0.00	Recession	-0.36	0.00
Problems	-0.19	0.00	Problems	-0.18	0.00
Convertible/preferred	-0.16	0.20	Convertible/preferred	-0.14	0.24
Record high	0.12	0.08	Record high	0.13	0.07
Options/VIX	-0.11	0.76	IPNCONGD (FRED-MD)	-0.10	0.98
Full R^2	0.25		Full R^2	0.25	
Benchmark R^2	0.02		Benchmark R^2	0.09	

Table II: reconstructing stock market returns

5.4 Table III + Table IV: financial activity and industry volatility

Read:

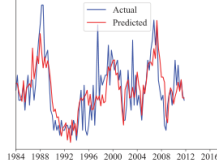
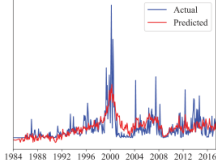
- Table III shows that topic attention reconstructs IPO volume, LBO volume, bankruptcy filings, and European CDS spreads.
- IPO volume is linked to IPO and venture capital topics; LBO volume is linked to takeovers and control stakes.
- Bankruptcy filings are strongly associated with recession attention.
- Table IV shows that industry-specific volatility is linked to thematically related topics such as automotive, banking, pharmaceuticals, computer hardware, oil and gas, and tobacco.

Economic message: The selected topics often match the economic meaning of the series, which supports the interpretability of the model.

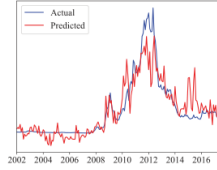
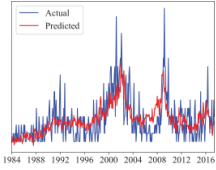
Table III
Reconstructing Financial Activity

Five-regressor active-set regression estimates based on lasso selection with p -values adjusted for postselection inference.

Topic	Coeff.	p -Val.	Topic	Coeff.	p -Val.
IPO Volume			LBO Volume		
IPOs	0.26	0.00	Takeovers	0.31	0.28
Venture cap.	0.22	0.00	Insurance	-0.28	0.02
Bankruptcy	-0.14	0.25	Control stakes	0.27	0.03
Internet	0.13	0.08	Key role	0.23	0.70
M&A	0.09	0.72	Job cuts	-0.18	0.13
R^2	0.44		R^2	0.58	



Bankruptcy Filings			European CDS Spreads		
Recession	0.48	0.00	Euro. sov. debt	0.58	0.00
Venture cap.	0.25	0.01	Govt. budgets	0.17	0.61
Accounting	0.24	0.01	Middle east	0.16	0.08
Small caps	0.12	0.00	Mobile devices	0.09	0.06
Machinery	-0.06	0.31	News conference	0.07	0.98
R^2	0.42		R^2	0.78	



from Ken French's website, and measure monthly volatility as the standard

Table III: financial activity

Table IV
Reconstructing Industry Volatility (Innovations)

Five-regressor active-set regression estimates based on lasso selection with p -values adjusted for postselection inference.

Topic	Coeff.	p -Val.	Topic	Coeff.	p -Val.	Topic	Coeff.	p -Val.
Automotive			Banking			Pharmaceuticals		
Automotive	0.16	0.03	Nonperform. loans	0.26	0.00	Clintons	0.20	0.01
Fees	0.14	0.11	Mortgages	0.20	0.00	News conf.	-0.14	0.48
Mutual funds	-0.13	0.33	Options/VIX	-0.18	0.00	Pharma	0.14	0.10
Corporate govt.	0.12	0.59	Chemicals/paper	-0.15	0.10	Earnings fcts.	0.12	0.35
Financial crisis	-0.08	0.87	NASD	-0.09	0.76	Health ins.	0.10	0.62
R^2	0.10		R^2	0.17		R^2	0.11	
Computer Hardware			Oil and Gas			Tobacco		
Retail	0.20	0.00	Oil market	0.34	0.00	Tobacco	0.17	0.34
Computers	0.17	0.01	Earnings losses	-0.15	0.05	Problems	-0.10	0.43
Software	0.10	0.18	Options/VIX	-0.15	0.87	Mexico	0.10	0.98
Fees	-0.09	0.72	Cultural life	0.14	0.08	Terrorism	-0.09	0.41
Russia	-0.08	0.93	Elections	-0.13	0.46	Lawsuits	0.09	0.59
R^2	0.09		R^2	0.21		R^2	0.08	

Table IV: industry volatility

5.5 Table V + Figure 4: interactions and macro dynamics

Read:

- Table V allows pairwise topic interactions.
- The selected predictors for industrial production and employment are interactions, many involving recession attention.
- Figure 4 shows that a recession-attention shock predicts a large decline in industrial production and employment.
- The baseline decline is about 1.99% for industrial production after 17 months and about 0.92% for employment after 20 months.
- Robustness checks change the ordering, include EPU, include Michigan sentiment, include VIX and trends, and add more lags.

Economic message: Recession news attention contains incremental information about future macroeconomic dynamics beyond standard numerical indicators.

Table V
Reconstructing Macroeconomic Time Series with Interactions

Five-regressor lasso regression estimates and fits with p -values are adjusted for postselection inference, using individual topic attention regressors as well as their pairwise interactions.

Industrial Production Growth			Employment Growth		
Topic	Coeff.	p -Val.	Topic	Coeff.	p -Val.
Recession $\times$ Oil market	-0.25	0.00	Inv. banks $\times$ Bush/Obama/Trump	-0.28	0.94
Inv. banks $\times$ Bush/Obama/Trump	-0.18	0.17	Recession $\times$ Fees	-0.22	0.24
Job cuts $\times$ Recession	-0.09	0.73	Publishing $\times$ Recession	-0.19	0.51
Recession $\times$ Corporate governance	-0.07	0.94	Recession $\times$ Bankruptcy	-0.17	0.04
Short sales $\times$ Recession	-0.02	0.11	Credit ratings $\times$ Recession	-0.07	0.60
R^2	0.24		R^2	0.63	

Table V: topic interactions

5.6 Figure 5 + Figure 6: stock market links and no-look-ahead robustness

Read:

- Figure 5 links recession attention to stock market dynamics.
- Output responds negatively to both recession attention and stock market value, but the recession-attention response is large and persistent.
- Stock prices respond more strongly to recession attention than to EPU in the plotted comparison.
- Figure 6 compares full-sample LDA with online LDA and shows very similar impulse responses.

Economic message: News attention is not merely a restatement of the stock market, and the results are not driven by look-ahead bias in the full-sample topic model.

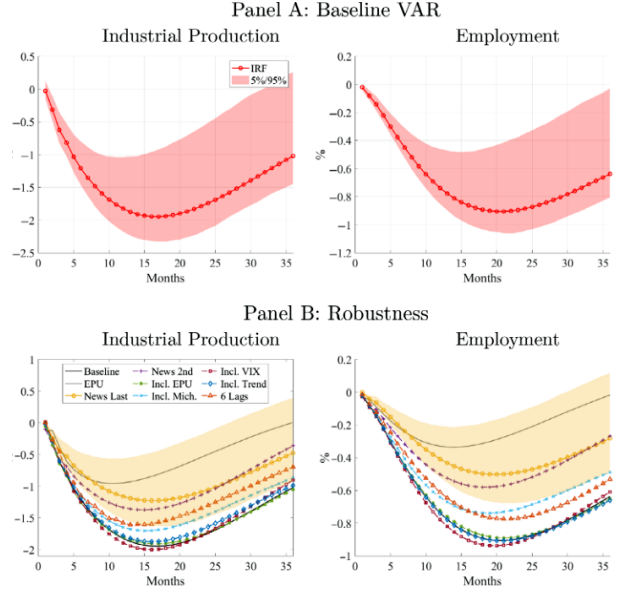


Figure 4: Output and employment responses to “recession” news attention shock. Responses of output and employment to a “recession” attention impulse in various VAR specifications, 1985 to 2017. The red shaded regions in the top plots are 90% confidence bands. This gold shaded regions in the bottom plots are 90% confidence bands when news attention is ordered last in the VAR (corresponding to the gold impulse-response curves). (Color figure can be viewed at rileyonlinelibrary.com)

Figure 4: output and employment responses

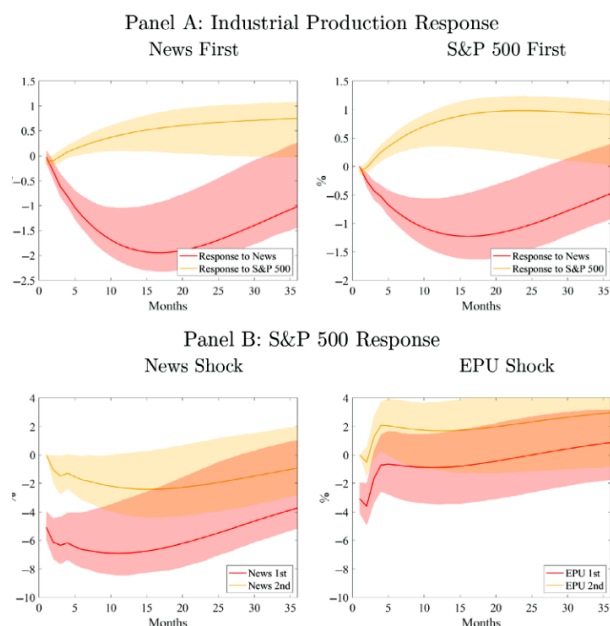


Figure 5. Stock market and "recession" attention. Panel A plots the response of output to impulses in "recession" attention and stock market value. Panel B reports the response of stock market value to an impulse in "recession" attention in our baseline VAR (left plot) and an impulse in EPU in the BBD VAR (right plot), and includes specifications for impulse variables ordered either before or after the S&P 500 in their respective VARs. Sample is 1985 to 2017. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 5: stock market and recession attention

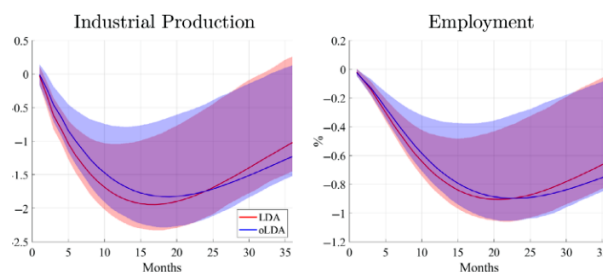


Figure 6. Responses to "recession" news attention shock: Avoiding look-ahead bias with online LDA. Responses of output and employment to a "recession" attention impulse for LDA versus oLDA, 1985 to 2017. The red lines represent the LDA point estimates and the red shaded regions 90% confidence bands. The blue lines represent the oLDA point estimates and the blue shaded regions 90% confidence bands. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 6: online LDA robustness

5.7 Figure 7 + Figure 8 + Figure 9: long sample and topic selection

Read:

- Figure 7 extends recession attention back to 1890 using front-page abstracts and shows high recession attention around NBER recessions.
- Figure 8 repeats the recession-attention VAR in the longer 1939–2017 sample.
- Short-run output and employment responses are similar across the old and recent samples, while long-run responses are more persistent in the longer sample.
- Figure 9 shows group-lasso VAR selection; cross-validation selects the recession topic as the only news attention topic in the VAR.

Economic message: Recession attention is robust across samples and is selected statistically, not just because the authors chose it manually.

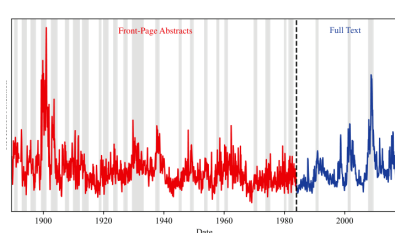


Figure 7. Recession attention and NBER recessions, 1890 to 2017. Time series of recession attention and NBER recessions, 1890 to 2017.

Figure 7: recession attention, 1890–2017

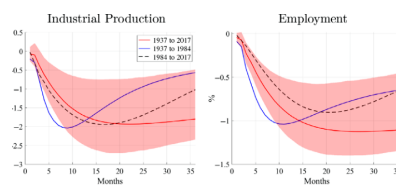


Figure 8. Responses to "recession" news attention shock: 1939 to 2017 sample. Responses output and employment to a "recession" attention impulse, 1939 to 2017. We estimate news attention over the 1939 to 1984 period by applying our rich topic model to the WSJ front-page abstracts used by Manella and Moreira (2017). The shaded regions are 90% confidence bands. (Color figure can be viewed at wileyonlinelibrary.com)

Figure 8: long-sample VAR

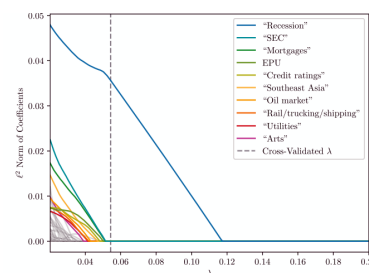


Figure 9: group-lasso selection

5.8 Figure 10: narrative retrieval

Read:

- Figure 10 plots one-month-ahead industrial production growth forecasts from the baseline VAR and the contribution from recession attention.

- The figure labels major forecast drops with the WSJ articles most representative of recession attention in those months.
- Examples include consumer confidence, high vacancies, retail sales, service sector orders, and global growth concerns.

Economic message: The method converts a statistical forecast into an interpretable set of narratives and articles.

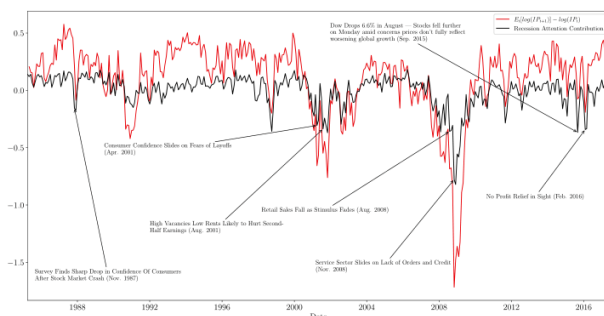


Figure 10. Narrative retrieval for industrial production growth forecasts. The figure shows one-month-ahead growth rate expectations from the baseline VAR (red line) and contribution to the expectation coming from "recession" attention (black line). We annotate seven of the largest shifts in "recession" attention-based expectations. Annotations are headlines of articles that have the greatest allocation to the "recession" topic in a given month. (Color figure can be viewed at wileyonlinelibrary.com)

5.9 Table VI + Figure 11: market-timing performance

Read:

- Table VI compares out-of-sample market-timing strategies.
- The LDA and oLDA strategies have Sharpe ratios of 0.99 and 1.04, compared with 0.71 for buy-and-hold.
- oLDA has the highest expected return, the strongest information ratio, and the smallest maximum loss among the news-based strategies.
- Figure 11 shows cumulative returns, with LDA and oLDA outperforming the benchmark strategies.
- Major outperformance episodes are linked to topics such as internet, private/public sector, and arts.

Economic message: Topic attention is not only descriptive. It contains out-of-sample information useful for market timing.

Table VI
Out-of-Sample Performance of Market-Timing Strategies

This table reports annualized performance measures for the different market-timing strategies. Buy-and-hold corresponds to a strategy that holds the market. LDA corresponds to a lasso strategy using our original full-sample LDA attention series. oLDA corresponds to a lasso strategy that uses oLDA in-place of full-sample LDA. R-word uses a prediction based on mentions of the word "recession" in the WSJ. EPU uses a prediction based on the EPU index. GW uses the 15 predictors from the Welch and Goyal (2008) "kitchen sink" regression. GW Lasso uses a similar lasso procedure as oLDA with the same 15 predictors from Welch and Goyal (2008). All strategies except buy-and-hold, hold the forecasted market return to the following month.

	Buy-and-Hold	LDA	oLDA	R-Word	EPU	GW	GW Lasso
Sharpe ratio	0.71	0.99	1.04	0.65	0.53	0.53	0.62
Expected return	2.72	3.79	4.01	2.50	2.04	2.03	2.40
t-Stat	2.69	3.76	3.97	2.48	2.02	2.01	2.37
Information ratio		0.29	0.31	-0.46	-0.50	-0.18	-0.07
α		2.30	2.87	-0.20	-0.51	0.61	1.55
α t-Stat		2.67	3.06	-1.54	-1.43	0.69	1.58
Max loss	-47.09	-47.31	-25.89	-46.98	-77.66	-65.76	-64.31
Skew	-0.56	0.20	1.04	-0.66	-1.41	-0.71	0.05

Table VI: market timing

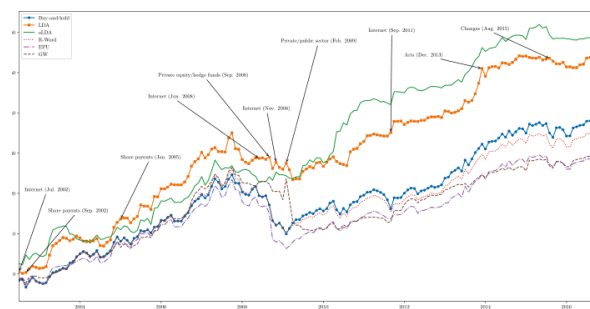


Figure 11. Cumulative returns for market-timing strategies. This figure plots the cumulative returns for different market-timing strategies over the testing sample. Buy-and-hold corresponds to the benchmark strategy. (Color figure can be viewed at wileyonlinelibrary.com)

6 Short summary

- The paper proposes a textual measurement system for business cycles using the full text of the *Wall Street Journal*.
- LDA summarizes roughly 800,000 articles into 180 interpretable topics.
- Monthly topic-attention series act as quantitative indices of business narratives.
- Topic attention reconstructs macroeconomic, financial, and industry-level time series.
- Recession attention is the most important topic for macroeconomic dynamics.
- In a text-augmented VAR, recession attention predicts large future declines in output and employment.
- Results are robust to VAR ordering, controls, longer samples, and online LDA.
- Group lasso selects recession attention as the relevant news topic for the VAR.
- Narrative retrieval links model forecasts back to specific WSJ articles.
- Topic attention improves out-of-sample market timing relative to common benchmark predictors.
- The paper’s main contribution is a bridge between high-dimensional machine learning and readable economic narratives.

7 Conclusion

7.1 Core conclusion

Bybee, Kelly, Manela, and Xiu show that business news text provides a rich and interpretable measure of the state of the economy. Topic attention summarizes how the newspaper allocates coverage across economic narratives, and these attention measures track and forecast macroeconomic and financial variables.

7.2 Economic intuition

News media aggregates dispersed information about events, expectations, and interpretations. Topic models compress this information into a manageable set of attention indices. Because these indices capture the narratives that households, firms, investors, and policymakers are reading, they contain information that is distinct from standard numerical time series.

7.3 Takeaway

The paper’s lasting message is that text can be used as structured economic data. A well-designed topic model can measure business-cycle narratives, forecast macro-finance outcomes, and retrieve the underlying articles that explain the forecasts.